

1 Solution — GIAM 6.6

Problem 1

1.1 Problem

The n -th triangular number is given by

$$T(n) = \frac{n^2 + n}{2}.$$

Regarded as a function $T : \mathbb{R} \rightarrow \mathbb{R}$, this fails the horizontal line test (it is a parabola) and so is not invertible. Find a suitable restriction of the domain on which T becomes invertible.

1.2 Setup — Why It Fails

Rewrite T as a parabola opening upward:

$$T(x) = \frac{1}{2}(x^2 + x) = \frac{1}{2}\left(\left(x + \frac{1}{2}\right)^2 - \frac{1}{4}\right).$$

The vertex is at $(x, T(x)) = \left(-\frac{1}{2}, -\frac{1}{8}\right)$, and T is symmetric about the vertical line $x = -\frac{1}{2}$:

$$T(a) = T(b) \iff a = b \text{ or } a + b = -1.$$

So every $y > -\frac{1}{8}$ has *two* distinct preimages — one to the left of the vertex and one to the right — and the function fails the horizontal line test on all of $\mathbb{R}$.

1.3 A Suitable Restriction

Restrict the domain to the right half-line $\left[-\frac{1}{2}, \infty\right)$. On this half:

- The derivative $T'(x) = x + \frac{1}{2} \geq 0$, with strict inequality for $x > -\frac{1}{2}$. Hence T is *strictly increasing*.
- The image is $\left[-\frac{1}{8}, \infty\right)$: the minimum $-\frac{1}{8}$ is attained at the endpoint $x = -\frac{1}{2}$, and $T \rightarrow \infty$ as $x \rightarrow \infty$.

So

$$T : \left[-\frac{1}{2}, \infty\right) \longrightarrow \left[-\frac{1}{8}, \infty\right)$$

is a bijection, and therefore invertible.

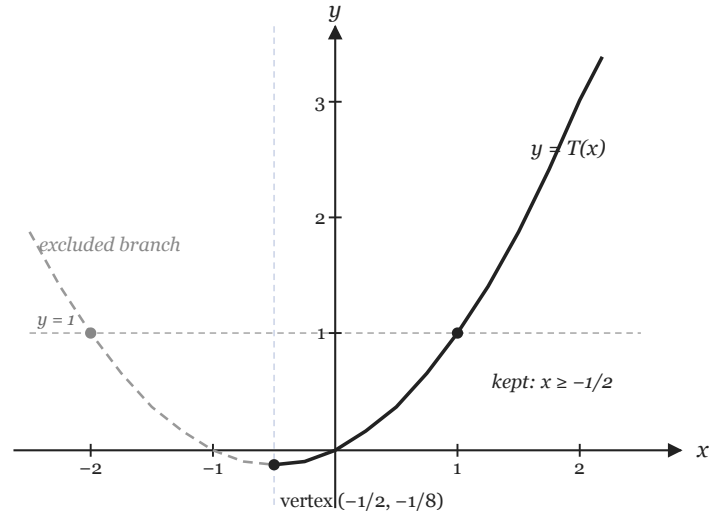


Figure 1.1: $T(x) = (x^2 + x)/2$ as a parabola. The dashed left branch ($x < -1/2$) is excluded; the solid right branch ($x \geq -1/2$) is the kept restriction. The light dashed line at $y = 1$ illustrates the horizontal-line-test failure on the full domain: it meets the curve at $x = -2$ (excluded) and $x = 1$ (kept).

Restricting to the right branch leaves only one preimage.

1.4 The Inverse

Solve $y = \frac{1}{2}(x^2 + x)$ for x . Multiplying through by 2 and rearranging,

$$x^2 + x - 2y = 0,$$

a quadratic in x . The quadratic formula gives

$$x = \frac{-1 \pm \sqrt{1 + 8y}}{2}.$$

On the restricted domain $x \geq -\frac{1}{2}$ we want the larger root, so the $+$ branch:

$$\boxed{T^{-1}(y) = \frac{-1 + \sqrt{1 + 8y}}{2}, \quad y \geq -\frac{1}{8}.$$

(Choosing the $-$ branch instead would give the inverse of T restricted to the *left* half $(-\infty, -\frac{1}{2}]$.)

1.5 Sanity Check

The recipe should send each non-negative integer triangular number back to its index:

y	$T^{-1}(y)$	Check
0	$\frac{-1+1}{2} = 0$	$T(0) = 0$
1	$\frac{-1+3}{2} = 1$	$T(1) = 1$
3	$\frac{-1+5}{2} = 2$	$T(2) = 3$
6	$\frac{-1+7}{2} = 3$	$T(3) = 6$
10	$\frac{-1+9}{2} = 4$	$T(4) = 10$
15	$\frac{-1+11}{2} = 5$	$T(5) = 15$

Table 1.1

All consistent.

1.6 Remark — Other Valid Restrictions

The restriction $[-\frac{1}{2}, \infty)$ is the *largest* domain on which T is increasing, but several alternatives also work:

- **Left branch.** Restrict to $(-\infty, -\frac{1}{2}]$; here T is strictly decreasing, with the same image $[-\frac{1}{8}, \infty)$. The inverse is the $-$ branch: $T^{-1}(y) = (-1 - \sqrt{1+8y})/2$.
- **Non-negative reals.** Restrict to $[0, \infty)$; T is strictly increasing here too (since the vertex sits at $-\frac{1}{2} < 0$), with image $[0, \infty)$. The inverse is the same $+$ branch formula but only valid for $y \geq 0$. This is the most natural restriction in the original setting, where the triangular numbers $T(0), T(1), T(2), \dots$ are an integer-indexed sequence.
- **Non-negative integers.** Restrict further to $\mathbb{Z}^{\text{noneg}}$. This recovers the classical sequence $0, 1, 3, 6, 10, 15, \dots$ as a bijection with its image — the set of triangular numbers — and the inverse is the integer-valued version of the formula above.

Each restriction trades scope for invertibility; the right-branch restriction $[-\frac{1}{2}, \infty)$ is the “largest possible” answer, while $[0, \infty)$ is the “natural setting” answer.